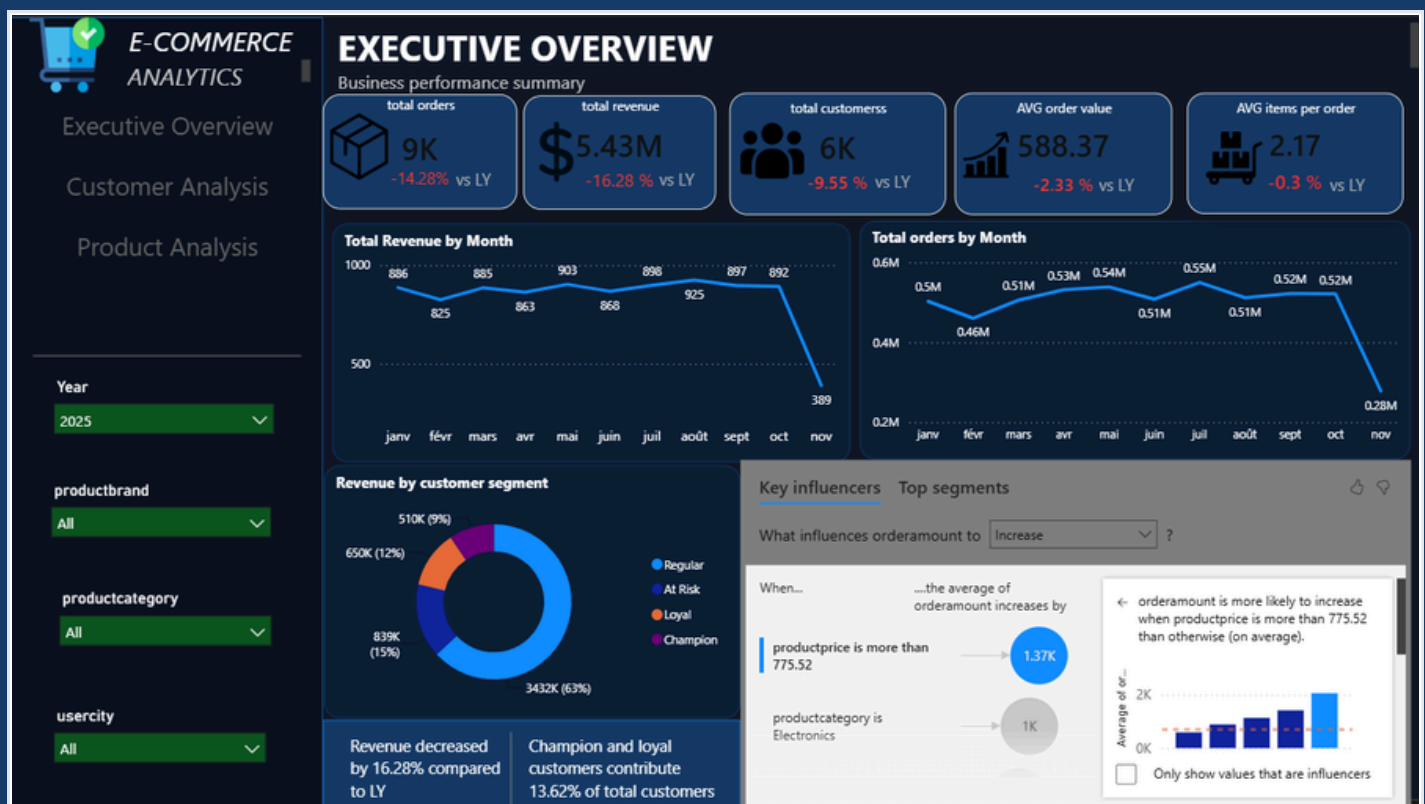


E-Commerce Business Intelligence

End-to-End Sales & Customer Analytics

From Raw Data to Business-Ready Insights



SQL SERVER · EXCEL · POWER QUERY · POWER BI · DAX

Adem Toumi

Business Intelligence Portfolio Project

Turning E-Commerce Data Into Decisions

The project started with six raw datasets covering customers, products, orders, order items, reviews, and behavioral events. None of it was analysis-ready.

The challenge was never just visualizing numbers. It was building a reliable workflow that could turn inconsistent, disconnected files into structured information — and eventually, into dashboards a business could actually act on.



PROJECT AT A GLANCE

Data domains	Customers · Orders · Products · Reviews · Events
Tools	Excel · Power Query · SQL Server · Power BI
Final output	3 interactive dashboards

Starting With the Data, Not the Dashboard

CSV Files



Excel Exploration



Power Query



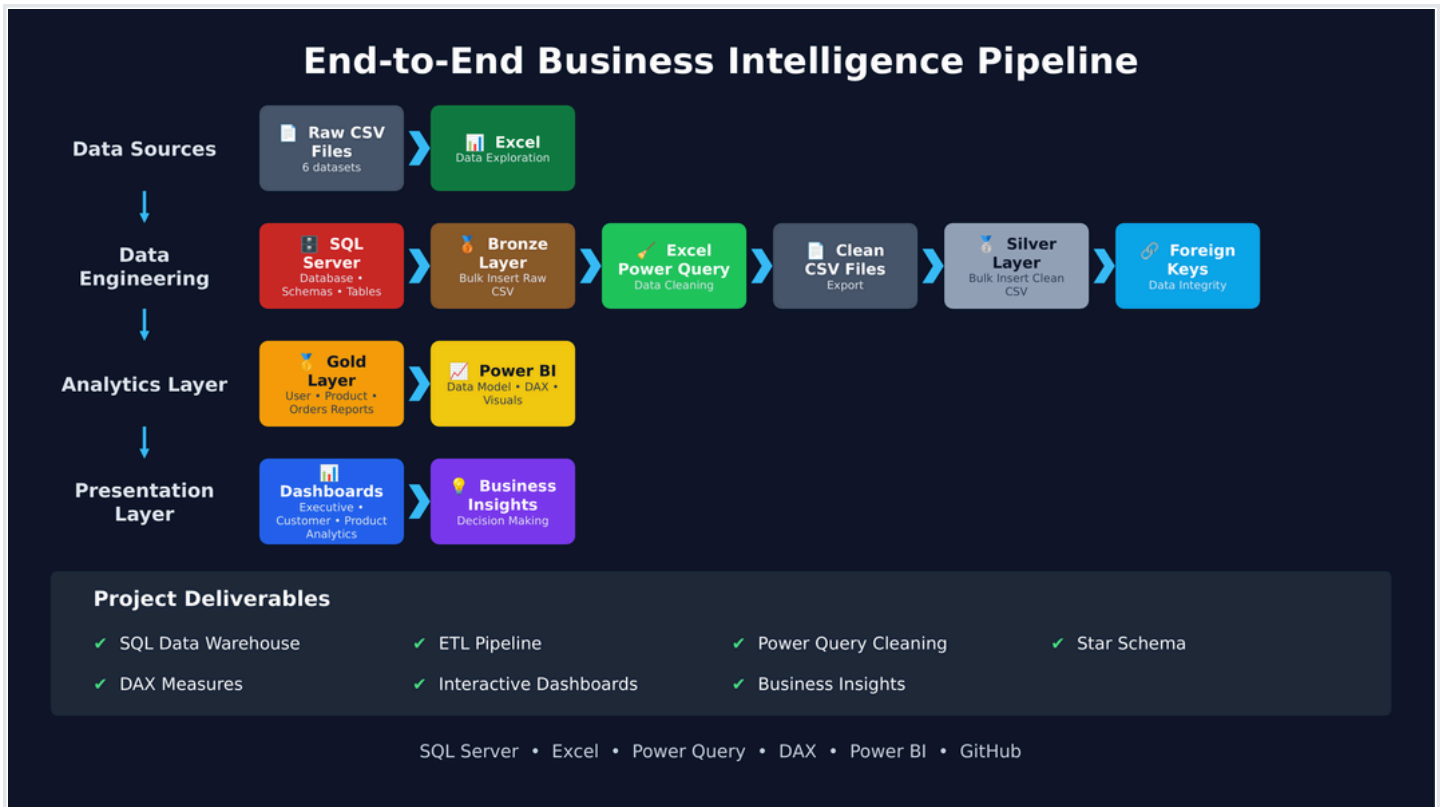
Clean CSV Files

	order_itemid	orderid	productid	userid	quantity	Itemprice	Itemtotal
1	14120	6501	494	8033	1	148.44	148.44
2	25853	11916	1174	375	1	30.81	30.81
3	19480	9007	638	5565	2	76.4	152.8
4	7365	3352	55	6874	1	949.73	949.73
5	23017	10621	72	5319	1	39.59	39.59
6	5378	2446	431	1162	2	27.08	54.16
7	39603	18157	201	7069	2	23.78	47.52
8	13182	6058	1922	4926	1	201.36	201.36
9	37484	17208	135	688	3	49.41	148.23
10	40288	18477	1447	571	1	26.07	26.07
11	15084	6952	1552	8359	3	27.47	82.41
12	5790	2626	1305	9369	1	864.19	864.19
13	14551	6705	456	3782	1	24.35	24.35
14	18613	8597	418	6469	1	1413.41	1413.41
15	24876	11480	1748	1356	1	56.34	56.34
16	27970	12865	177	7628	1	17.35	17.35
17	2569	1180	1769	7176	1	18.76	18.76
18	7620	3470	302	132	1	47.73	47.73
19	39700	18207	1757	8488	1	206.86	206.86
20	30551	14036	891	4927	3	29.68	89.04
21	7655	3489	1271	3410	1	83.6	83.6
22	36723	16851	969	9717	1	12.65	12.65
23	41644	19106	1914	4617	1	948.69	948.69
24	29100	13390	1284	5969	2	153.71	307.42
25	2434	1121	1427	2918	1	238.36	238.36
26	38293	17551	351	5792	2	84.8	169.6
27	32693	15008	1506	3219	3	186.05	558.15
28	31455	14445	1067	9971	1	98.53	98.53

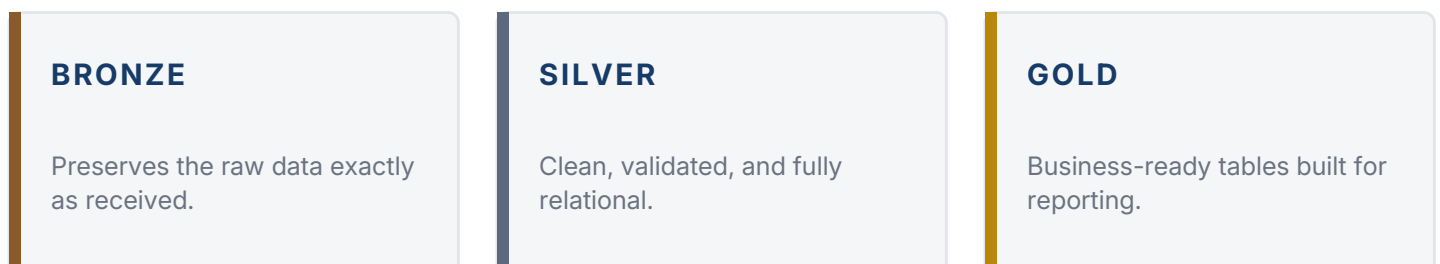
Every raw table was explored in Excel before any cleaning began — checking structure, data types, and inconsistencies. The order items table shown above, for example, needed several rounds of column splitting and renaming before its fields lined up cleanly.

Once a table's issues were understood, Power Query handled the actual cleaning: fixing types, removing duplicates, and standardizing formats. Each cleaned table was then exported as a fresh CSV, ready to be loaded into the Silver layer.

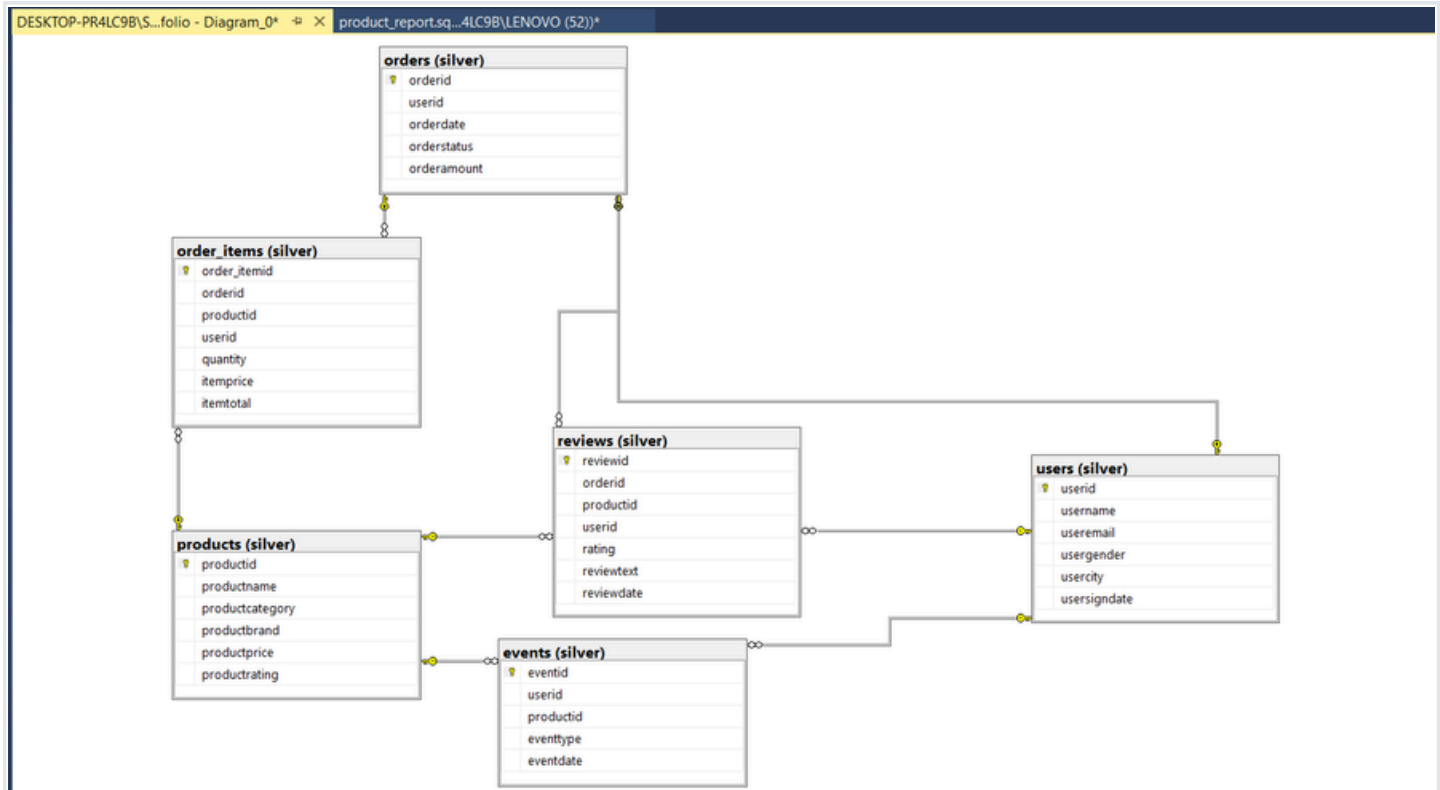
From Raw Data to a Business-Ready Warehouse



The warehouse follows a layered design, commonly known as the Medallion architecture, where each layer has exactly one job.



Building Relationships That Can Be Trusted



Once cleaned, the six Silver-layer tables were explicitly connected through eight foreign key relationships in SQL Server — turning a set of loose tables into a fully relational, referentially sound database.

- Orders → Users
- Order Items → Orders
- Order Items → Products
- Reviews → Orders
- Reviews → Users
- Reviews → Products
- Events → Users
- Events → Products

Preparing the Data for Analysis



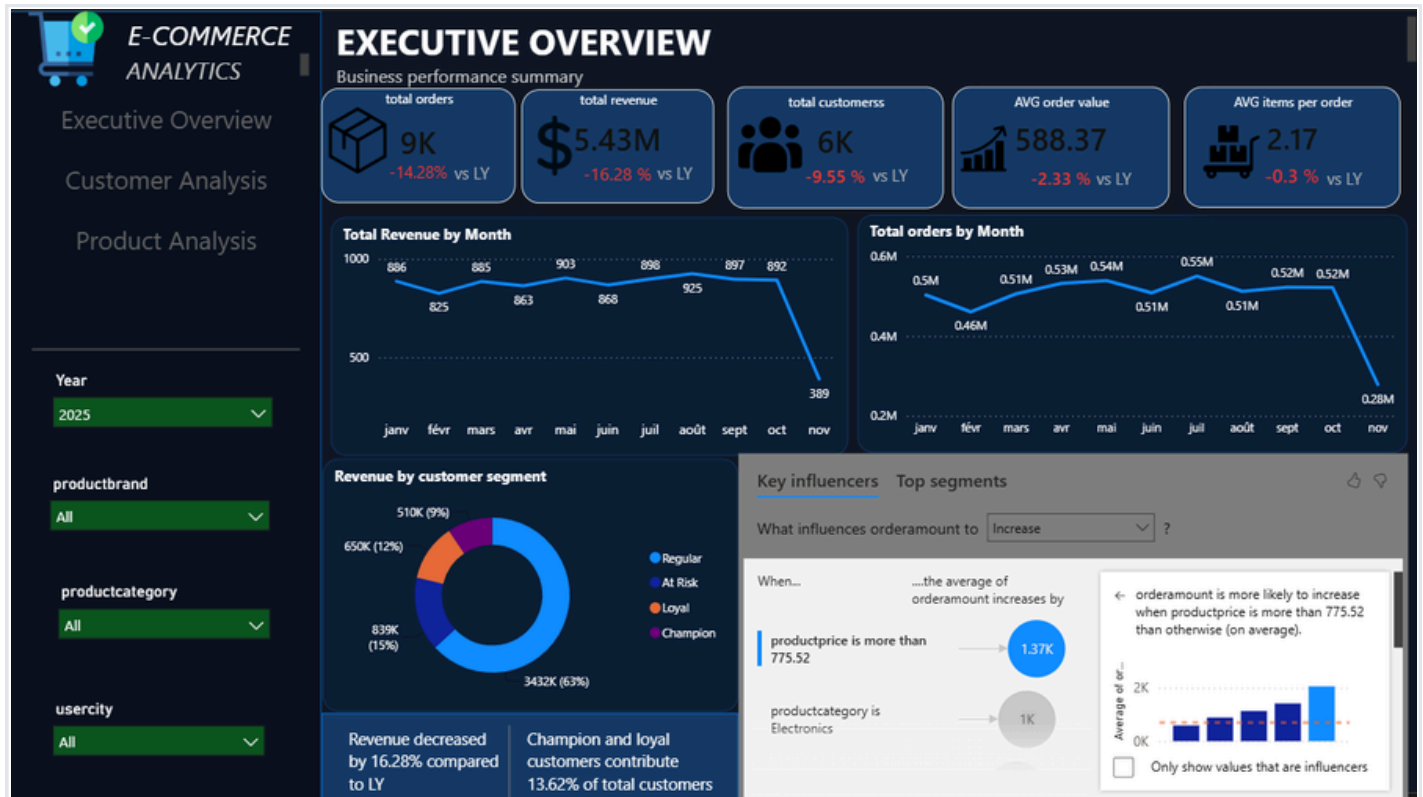
Rather than exposing raw transactional tables to Power BI, three Gold-layer tables — **user_report**, **product_report**, and **orders_report** — were built specifically for reporting, alongside a dedicated date table for time intelligence.

Some calculations still needed to respond dynamically to filters and dates — total customer spend, for instance, was better handled as a DAX measure than as a static column:

```
Customer Total Sales =  
SUMX(  
    VALUES(orders_report[orderid]),  
    CALCULATE(MAX(orders_report[orderamount]))  
)
```

Business logic lives in SQL. Dynamic, filter-aware calculations live in DAX.

One View of the Business



PURPOSE

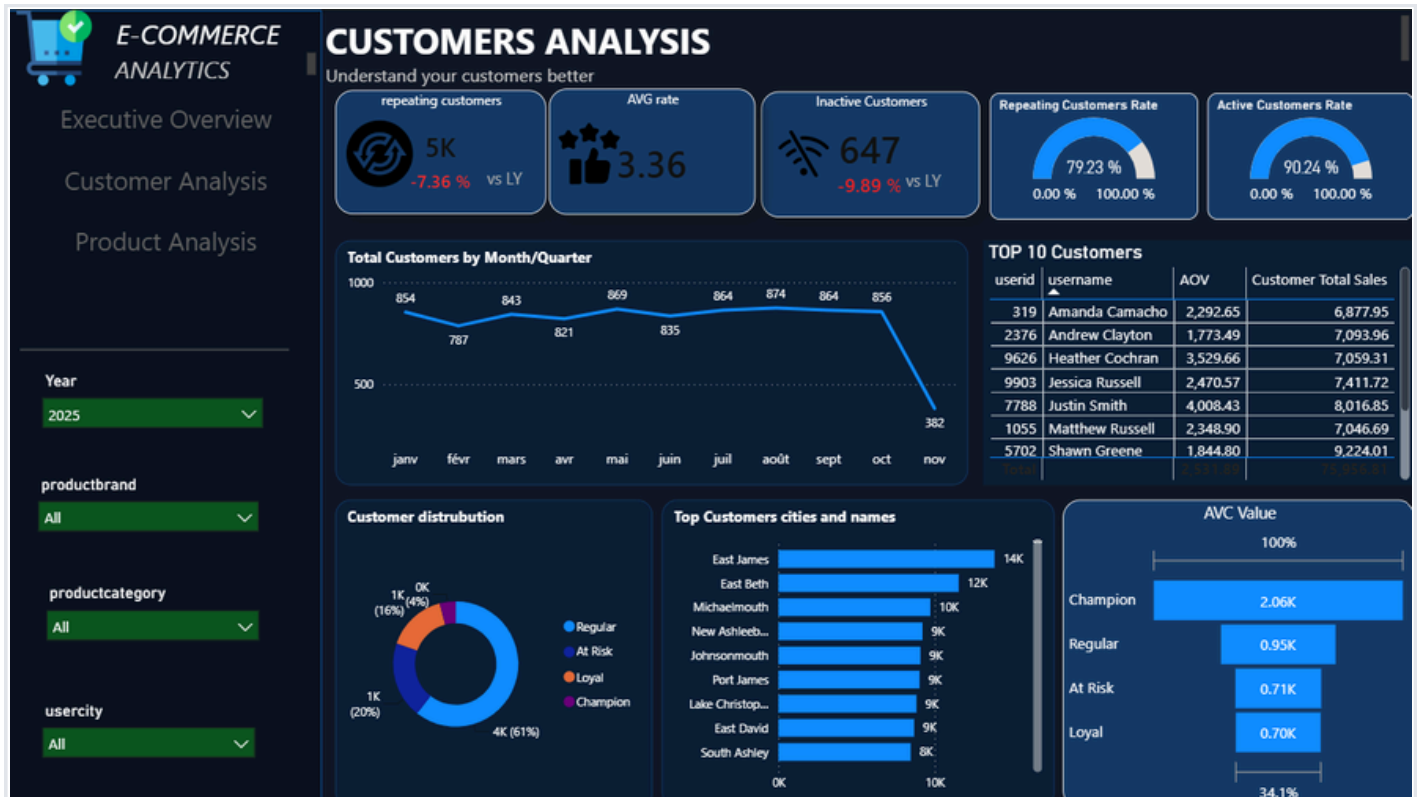
A single entry point for decision-makers to gauge overall business health at a glance.

KEY AREAS

Orders, revenue, and customers vs. last year — plus segment mix and an AI-driven view of what's moving order value.

- Revenue reached \$5.43M, down 16.28% year over year, on 9K total orders.
- The Regular segment drives 63% of revenue, while Champion and Loyal customers together make up just 13.62% of the customer base.
- The AI key-influencers panel links higher order value to Electronics purchases above \$775 in price.

Understanding Customers Beyond the Transaction



PURPOSE

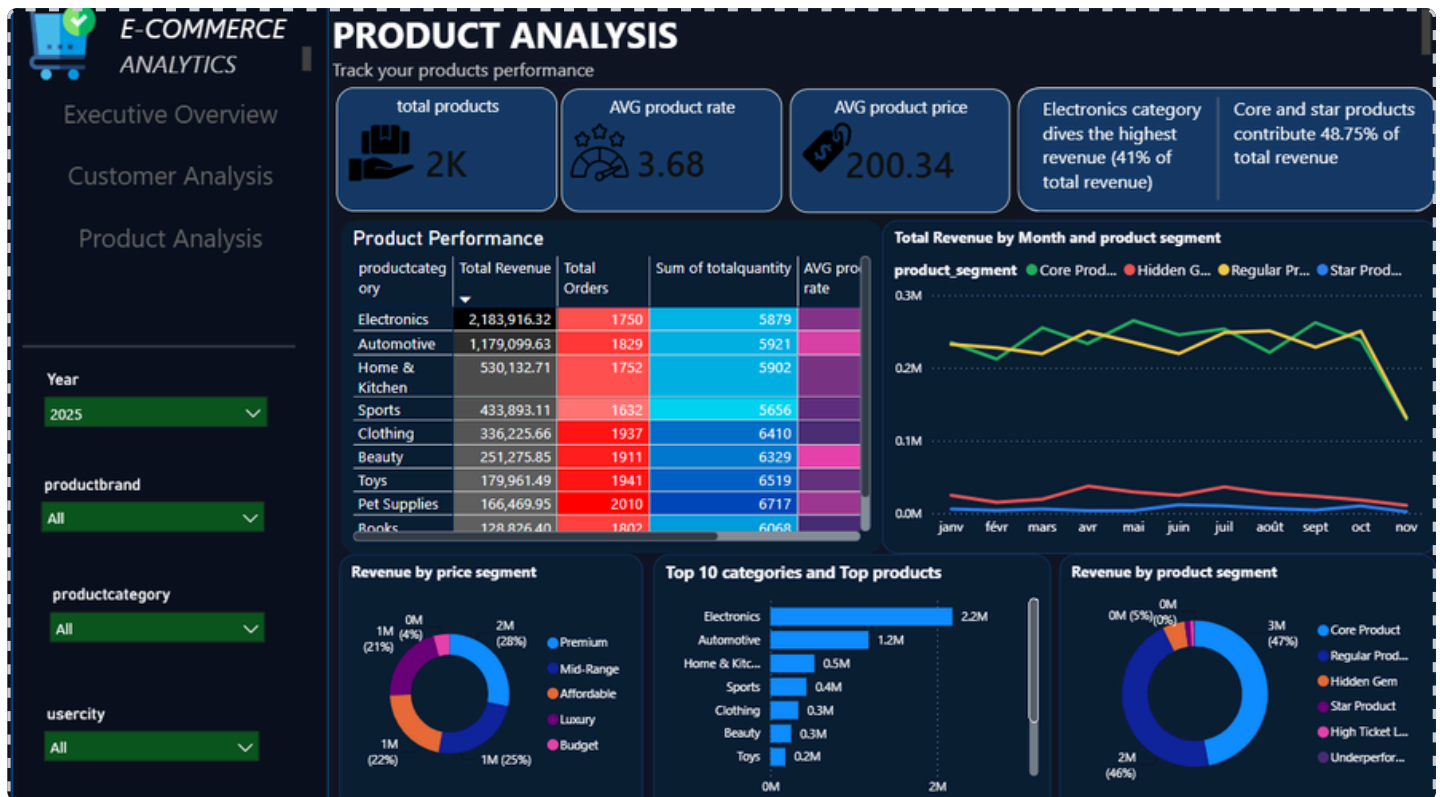
Move past raw customer counts and into who is actually valuable, active, or at risk of leaving.

KEY AREAS

RFM-based segmentation, repeat purchase and activity rates, and value comparison across segments.

- 79.23% of customers are repeat buyers, and 90.24% are currently active — healthy retention overall.
- 647 customers are already inactive, a concrete, addressable group for re-engagement.
- Champion customers carry roughly 2–3x the average value of Regular customers (2.06K vs 0.95K), making them worth protecting specifically.

Seeing Which Products Drive Performance



PURPOSE

Identify which categories and product segments actually move the business, beyond a simple top-sellers list.

KEY AREAS

Category performance, revenue by price tier, and revenue by strategic product segment.

- Electronics alone accounts for 41% of total revenue — more than double the next-largest category.
- Core and Star products together contribute 48.75% of total revenue, out of roughly six strategic segments.
- Revenue is concentrated in Mid-Range and Affordable price tiers, with Premium contributing a smaller but meaningful share.

What the Data Revealed

01

CUSTOMER VALUE CONCENTRATION

Champion and Loyal customers are just 13.62% of the customer base, yet Champions carry roughly double the average value of Regular customers.

WHY IT MATTERS

A small, identifiable group has outsized influence on revenue stability — worth protecting specifically, not just growing broadly.

02

REVENUE CONCENTRATION

Electronics alone drives 41% of total revenue — more than double the next category, Automotive.

WHY IT MATTERS

That concentration is an opportunity, but also a single point of exposure if demand shifts.

03

RETENTION, WITH A CLEAR GAP

79.23% of customers are repeat buyers and 90.24% are active, but 647 customers have already gone inactive.

WHY IT MATTERS

Overall retention is strong, which makes the inactive group a concrete, addressable target rather than a vague concern.

04

A FOCUSED PRODUCT CORE

Core and Star products contribute 48.75% of total revenue, out of the full product segment mix.

WHY IT MATTERS

Knowing exactly which products fall into these two segments should directly guide inventory and marketing priority.

From Raw Data to Decision-Ready Intelligence



TECHNICAL OUTCOME

A structured, layered BI workflow spanning Excel, SQL Server, and Power BI — from raw CSV files to a governed data warehouse with enforced relationships.

BUSINESS OUTCOME

Three interactive dashboards that let executives, and customer and product teams, each answer their own questions from the same trusted data.

FULL PROJECT REPOSITORY

github.com/ademtoumi-tn/E-Commerce-End-to-End-Business-Intelligence-Project

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